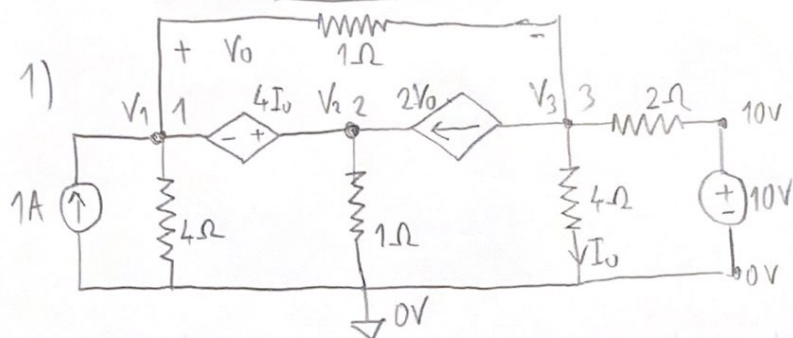


EE 201 - Assignment #1

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* We will use node voltage method and choose lower side of the circuit as reference node with 0 potential. Define super node 1-2 because we have a dependent voltage source between them.

$$\text{KCL (1-2)}: \frac{V_1 - V_3}{1} + \frac{V_1}{4} + \frac{V_2}{1} - 1 - 2V_0 = 0 \quad (1)$$

$$\text{KCL (3)}: \frac{V_3 - V_1}{1} + \frac{V_3 - 10}{2} + \frac{V_3}{4} + 2V_0 = 0 \quad (2)$$

And we have the following auxiliary conditions:

$$\frac{V_3}{4} = I_0, \quad V_2 - V_1 = 4I_0 = V_3 \quad (3), \quad V_1 - V_3 = V_0 \quad (4)$$

Using (1) and (4): $\nearrow V_1 + V_3$ use (3) $\times 4$

$$4V_1 - 4V_3 + V_1 + 4V_2 - 4 - 8V_1 + 8V_3 = 0 \implies V_1 + 8V_3 = 4$$

Using (2) and (4)

$$4V_3 - 4V_1 + 2V_3 - 20 + V_3 + 8V_1 - 8V_3 \implies 4V_1 - V_3 = 20$$

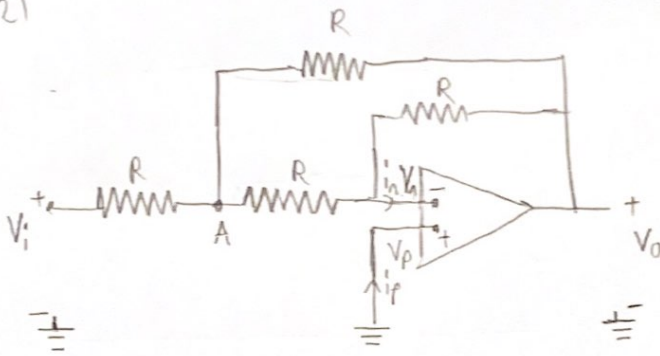
$$V_1 = \frac{164}{33} \text{ V}, \quad V_2 = \frac{160}{33} \text{ V}, \quad V_3 = \frac{-4}{33} \text{ V}$$

$$33V_3 = -4 \implies V_3 = \frac{-4}{33} \text{ V}$$

$$V_1 = \frac{20 + \frac{4}{33}}{4} = 5 - \frac{1}{33} = \frac{164}{33} \text{ V}$$

$$V_2 = \frac{160}{33}$$

2)



For an OPAMP :

$$V_n = V_p \quad \text{and} \quad i_n = i_p = 0$$

First we see that $V_A = V_p = 0$. So using KCL:

$$\text{KCL (n)} : \frac{V_A}{R} + \frac{V_o}{R} - i_n = 0 \Rightarrow V_A = -V_o$$

$$\text{KCL (A)} : \frac{V_A - V_i}{R} + \frac{V_A}{R} + \frac{V_A - V_o}{R} = 0 \Rightarrow V_i = -4V_o$$

$$\boxed{V_o = \frac{-V_i}{4}}$$